

Implement Threat Intelligence Principles

OpenCTI from GitHub + IOC analysis + vulnerability-risk integration for DeRexi

INSTRUCTOR CLARIFICATION

This revised version uses the official OpenCTI GitHub project as the threat-intelligence foundation. It also explains why vulnerability scanning matters, what DeRexi should implement after scan findings are identified, and how the risk component benefits the project. The existing IOC analysis is retained.

1. OpenCTI from GitHub - how DeRexi would use it

OpenCTI is an open-source cyber threat intelligence platform that organizes technical and non-technical threat information using STIX-based data. Its official GitHub ecosystem includes the core OpenCTI platform and a separate connectors repository used to ingest, enrich, or export threat-intelligence data.

- DeRexi use case: centralize threat indicators, vulnerabilities, threat actors, attack techniques, and related evidence instead of keeping the information in separate notes or tools.
- Implementation source: official OpenCTI-Platform/opencti and OpenCTI-Platform/connectors repositories on GitHub.
- Connector strategy: use the CVE connector for vulnerability intelligence and the MITRE ATT&CK connector for adversary behavior and technique context.
- If deployed later, connectors require an OpenCTI URL and dedicated connector token; the connectors can be run through Docker or configured manually according to the official documentation.

2. Two IOC examples already available to DeRexi

IOC type	SHA-256 file hash
Value	02aac4d201886e01abd2e3d42a20edcb4e6c2381549de19df0b9166607cd3f51
Detection	Endpoint/EDR hash matching, SIEM searches, and threat-intelligence lookup.
Threat relevance	Identifies the malicious Linux shell-script dropper analyzed in the earlier malware dossier.

IOC type	Malicious download infrastructure
Value	160.119.69.4 / http://160.119.69.4/x
Detection	Firewall, proxy, DNS, IDS/IPS, and outbound-connection monitoring.
Threat relevance	A DeRexi endpoint contacting this infrastructure could indicate payload staging or active compromise.

3. Why vulnerability scanning matters

A vulnerability scan gives DeRexi evidence about what systems are reachable, which services are exposed, and where configuration or known-vulnerability risk may exist. It converts assumptions about security into findings that can be reviewed, prioritized, and remediated.

Existing DeRexi scan evidence	Observed result	Why it matters
Discovery scope	10.11.2.0/24; 10 active hosts	Establishes the reachable attack surface before deeper testing.
Selected server	10.11.2.144	Treats a critical DeRexi application server as the prioritized asset.
Service exposure	SSH on TCP/22; no confirmed target-specific vulnerability	Shows that even a legitimate service can create risk if access, patching, or monitoring is weak.

From Scan Finding to Risk Decision

How threat intelligence turns DeRexi vulnerability evidence into practical security action

4. Implementation - what DeRexi should put in place

1. Scan and inventory	Continue authorized discovery and vulnerability scans to identify reachable systems, exposed services, and version information.
2. Enrich findings in OpenCTI	Use the CVE connector to add known-vulnerability intelligence and the MITRE ATT&CK connector to add adversary technique context.
3. Prioritize by business impact	Treat findings on authentication, policy data, remote administration, and core application services as higher priority because DeRexi supports financial-sector clients.
4. Apply controls	Restrict SSH to authorized administrators and trusted management networks, prefer strong/key-based authentication with additional controls, keep systems patched, and disable unnecessary services.
5. Monitor and re-scan	Review authentication and network telemetry for suspicious activity, then repeat scans to verify that remediation reduced exposure.

5. How DeRexi benefits from the risk component

Better prioritization: Threat intelligence helps distinguish a low-level exposure from a vulnerability that is actively exploited or tied to known attack behavior.

Reduced attack surface: Scan results identify services and systems that can be restricted, hardened, patched, or removed when they are unnecessary.

Protection of CIA: Risk decisions can be tied directly to confidentiality of policy/identity data, integrity of policy content and configuration, and availability of employee-facing services.

Faster incident decisions: Known IOCs, CVEs, and ATT&CK context give analysts clearer evidence when deciding whether activity should be investigated or escalated.

Financial-sector readiness: A repeatable scan -> intelligence -> remediation -> verification process gives DeRexi stronger evidence that security controls are being managed deliberately for a banking/finance environment.

RISK CONNECTION

The scan does not automatically equal risk. DeRexi uses the scan to identify exposure, OpenCTI to add threat and vulnerability context, and business impact to decide what must be fixed first. That is the value of the risk component: it turns technical findings into defensible security priorities.

6. Assignment criteria check

Official OpenCTI GitHub source used	YES - core platform and connectors repository documented.
2 IOCs analyzed	YES - malicious file hash + download infrastructure.
2 relevant connectors	YES - CVE + MITRE ATT&CK.
Importance of vulnerability scanning	YES - attack-surface discovery and evidence-based risk identification explained.
Implementation recommendations	YES - scan, enrich, prioritize, control, monitor, re-scan workflow.
Benefits for DeRexi	YES - risk prioritization, CIA protection, attack-surface reduction, faster decisions, financial-sector readiness.

SOURCE BASIS Official OpenCTI GitHub platform/connectors documentation; prior DeRexi Nmap vulnerability assessment; prior Malware Analysis Threat Dossier.